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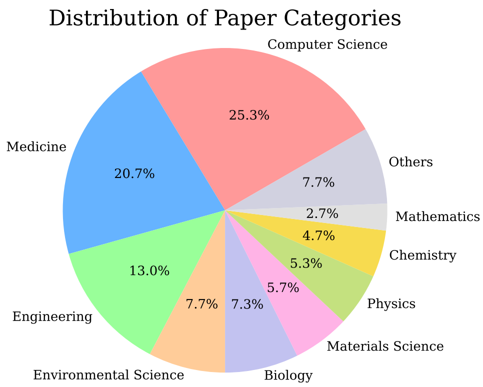

Figure 7: Visualization of the distribution of disciplines for all core papers, selected for research idea generation.

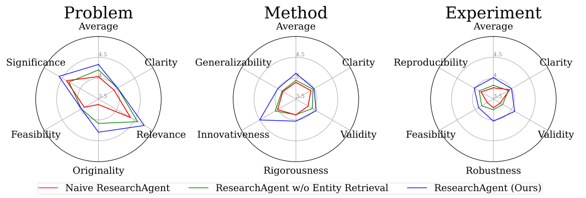

Figure 8: Results on our research idea generation task with model-based evaluation, where we exclude refinement steps.

A Additional Experimental Details

In this section, we provide additional details on experiments, including datasets, human evaluation setups, prompts (used for research idea generation and validation), and human-induced criteria.

A.1 Data Statistics

We visualize a distribution of core paper categories used for idea generation in Figure 7, where the categories are obtained from Semantic Scholar API[10]. From this, we find that the top 3 categories are computer science, medicine, and engineering.

A.2 Details on Human Evaluation

To conduct evaluations with human judges, we recruited 10 researchers from the United States and South Korea, majoring in computer science, medicine, and biology, each with a minimum of 3 published papers. For annotation, they were provided with a 6-page guideline document, which includes the task instruction and annotation examples. After reading this document, the annotators access the Label Studio platform, on which they first read the title and abstract of the target paper, and then review and evaluate the generated research ideas from different models. During the evaluation process, they are allowed to use any external tools, such as web searches. We note that they were compensated at a rate of $22.20 per hour. Also, on average, for one hour, they evaluated 3 sets of research ideas (that are generated from their own papers), with each set comprising three sub-ideas (the problem, method, and experiment design) from three different approaches (i.e., a total of 9 ideas for one hour). We perform three rounds of human evaluations with refinements in between, and, due to the cost associated with human annotations, we are able to fully evaluate a total of 150 ideas.

[10]https://www.semanticscholar.org/product/api

A.3 Prompts for Ideas Generation

We provide the prompts used to elicit the idea generations from our full ResearchAgent, specifically for instantiating problem identification, method development, and experiment design in Table 6, Table 7, and Table 8, respectively.

A.4 Prompts for Idea Validation

We provide the prompts used to elicit the idea validation from our ReviewingAgents as well as the model-based evaluations, specifically for instantiating problem validation, method validation, and experiment design validation in Table 9, Table 10, and Table 11, respectively. In addition, we provide the criteria used, which are induced by human judgments in the next subsection (Appendix A.5).

A.5 Criteria Induced by Human Judgements

Recall that, to align model-based evaluations with human preferences, we induce the criteria (used for automatic evaluations) with actual human judgments. We note that this is done by prompting GPT-4 with 10 pairs of generated ideas and (randomly selected) human judgments. We provide the resulting criteria for validations of problems, methods, and experiment designs in Table 13, Table 14, and Table 15, respectively.

B Additional Experimental Results

We provide additional experimental results, including comparisons without refinements and examples of the generated research ideas.

B.1 Results without Refinement Steps

To see whether the proposed ResearchAgent is consistently effective even without ReviewingAgents, we show the model-based evaluation results without any refinement steps in Figure 8. From this, we clearly observe that the full ResearchAgent outperforms its variants, demonstrating its effectiveness.

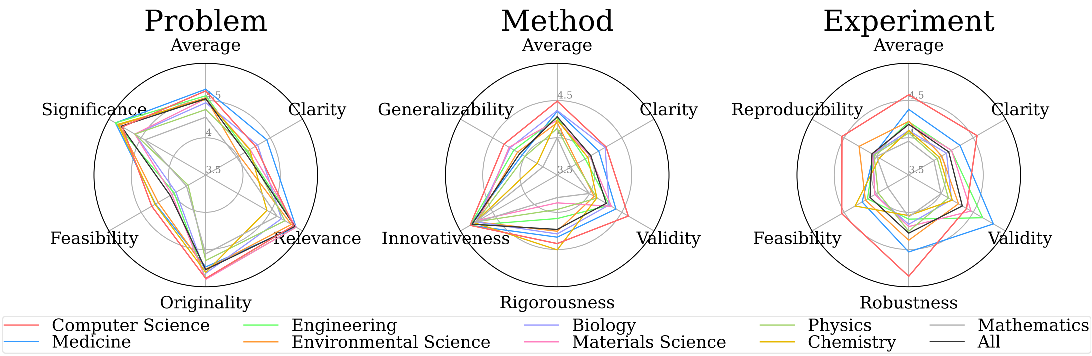

Figure 9: Breakdown results of the research ideas generated from our full ResearchAgent across different domains.

Table 5: Results with different entity retrieval strategies.

Methods	Problem	Method	Experiment
ResearchAgent			
- w/ Co-occurrence-based Retrieval	**4.52**	4.28	**4.18**
- w/ Embedding-based Retrieval	4.49	**4.34**	4.16
- w/o Entity Retrieval	4.35	4.13	4.02

B.2 Results on Generated Ideas by Domain

To see the quality of the generated research ideas across different domains, we breakdown the performance of ResearchAgent according to the categories of core papers in Figure 7, and present the results in Figure 9. From this, we observe that the generated research ideas on the high-resource domains (such as Computer Science, Medicine, and Engineering where there is a greater volume of existing literature as shown in Figure 7) are superior to those generated from the low-resource domain papers (such as Physics, Chemistry, and Mathematics). This disparity might be attributed to the fact that the underlying LLMs used to generate research ideas are likely trained on data predominantly sourced from high-resource domains, which leads to enhancing their ability to comprehend scientific concepts and produce relevant research ideas.

B.3 Analysis with Different Entity Retrieval

To see the effectiveness of different entity retrieval strategies, we perform additional experiments, replacing the co-occurrence-based entity retrieval in Equation 2 to the contextual embedding-based retrieval. Notably, this contextual embedding-based retrieval approach uses the entities that have the highest similarity to the entities appearing in the current literature (i.e., core paper and its references) used for idea generation, where the similarities are calculated based on embedding-level similarities between entities over the latent space represented by the entity linker (Wu et al., 2020). Therefore, unlike the previous co-occurrence-based entity retrieval that may retrieve entities that have opposite concepts to the main idea of the current core paper (since we often mention limitations of previous work along with the proposed ideas), this embedding-based approach may provide the ResearchAgent with mostly the entities having similar concepts to the core paper. Nevertheless, as shown in Table 5, the results with the strategy of entity co-occurrence-based retrieval are comparable to the results with the new embedding-based contextual retrieval. These results might confirm that there is not much difference in the quality of entity retrieval among those two strategies, i.e., most entities retrieved from the co-occurrence-based retrieval are contextually relevant for generating research ideas.